SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Assesment

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	K Sreelatha	**Reg. No.:**	192425248
Section / Batch:	AI & ML	**Date:**	20-07-26

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q1. ***Asymptotic Analysis of Matrix Multiplication Algorithms***

Analyze the efficiency of standard matrix multiplication using asymptotic notations. Clearly define the problem of multiplying two n × n matrices and identify the number of operations involved. Apply Big-O, Big-Theta, and Big-Omega notations to represent complexity in different cases. Use tabular steps or diagrams to illustrate operation growth. Present the findings clearly and justify the efficiency for large matrix sizes.

Code of Conduct

I __K Sreelatha___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Sree.K__

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.

Asymptotic Analysis of Matrix Multiplication

CSA0614 – Design Analysis And Algorithms

PRESENTATION BY:

k Sreelatha(192425248)

SUPERVISIOR:

Dr.A.Sairam

Problem Statement

- Matrix multiplication is a fundamental operation in computer science used in graphics, machine learning, scientific computing, and data analysis.
- Given two square matrices **A** and **B** of size **n** × **n**, determine the computational complexity of multiplying them using the standard matrix multiplication algorithm.
- The objective is to analyze the algorithm's efficiency using **Big-O**, **Big-Theta**, and **Big-Omega** asymptotic notations.
- As the matrix size increases, the number of arithmetic operations grows rapidly, making it essential to analyze the algorithm's time complexity and scalability for efficient performance.

Algorithm Overview

- Multiplies two **n × n** matrices to produce a result matrix **C**.
- Uses **three nested loops** to compute each element of the result matrix.
- Each element is calculated as the sum of products of corresponding row and column elements.
- Performs multiplication and addition operations systematically.
- Simple, easy to implement, and suitable for small to medium-sized matrices.
- **Input**
- Matrix A (n × n)
- Matrix B (n × n)
- **Output**
- Matrix C (n × n)
- **Formula**
- $C[i][j] = \sum_{k=1}^{n} A[i][k] \times B[k][j]$
- **Tip:** Place a small diagram on the right side of the slide:
- Matrix A × Matrix B
- (n × n) (n × n)
- ↓
- Matrix Multiplication
- ↓
- Matrix C (n × n)

Algorithm/Pseudo Code

Algorithm

- Read matrices **A** and **B** of size **n** × **n**.
- Initialize all elements of matrix **C** to **0**.
- Use three nested loops:
  - Traverse each row of **A**.
  - Traverse each column of **B**.
  - Multiply corresponding elements and add them to compute **C[i][j]**.
- Display the resultant matrix **C**.

Pseudocode

- for i = 0 to n-1
-     for j = 0 to n-1
-         C[i][j] = 0
-         for k = 0 to n-1
-             C[i][j] = C[i][j] + A[i][k] × B[k][j]
- return C

Working Example

Matrix A

1	2
3	4

Matrix B

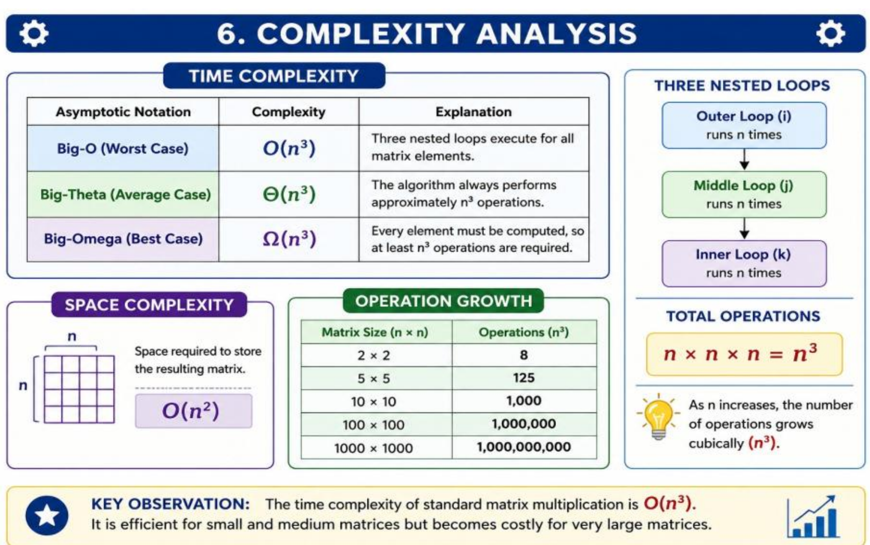
5	6
7	8

Calculation

$C_{11} = (1 \times 5) + (2 \times 7) = 19$
$C_{12} = (1 \times 6) + (2 \times 8) = 22$
$C_{21} = (3 \times 5) + (4 \times 7) = 43$
$C_{22} = (3 \times 6) + (4 \times 8) = 50$

Result Matrix C

19	22
43	50



7. COMPARATIVE ANALYSIS

COMPARISON OF MATRIX MULTIPLICATION ALGORITHMS

ALGORITHM	TIME COMPLEXITY	ADVANTAGES	DISADVANTAGES
STANDARD MATRIX MULTIPLICATION	$O(n^3)$	• Simple and easy to implement	• Slower for very large matrices
STRASSEN'S ALGORITHM	$O(n^{2.81})$	• Faster than the standard algorithm	• More complex and higher memory usage
COPPERSMITH–WINOGRAD ALGORITHM	$O(n^{2.376})$	• Efficient for extremely large matrices	• Difficult to implement in practice

KEY COMPARISON

1. **Standard Algorithm**
   Best for small and medium-sized matrices.
2. **Strassen's Algorithm**
   Reduces the number of multiplications, improving performance.
3. **Coppersmith–Winograd**
   Provides the lowest theoretical time complexity but is rarely used in practical applications due to its complexity.

Applications

🤖 **Machine Learning & Artificial Intelligence**

Used in training neural networks and performing deep learning computations.

🖼️ **Computer Graphics**

Enables 2D/3D transformations such as rotation, scaling, and animation.

📷 **Image Processing**

Supports image enhancement, filtering, compression, and feature extraction.

📊 **Scientific Computing**

Solves complex mathematical models and engineering simulations.

🤖 **Robotics & Automation**

Assists in robot motion planning, localization, and control systems.

🔐 **Cryptography**

Plays a role in encryption and secure data transmission.

🌐 **Data Analysis**

Used in recommendation systems, graph algorithms, and large-scale data processing

Result & Discussion

Results

Matrix Size (n × n)	Time Complexity	Operation Growth
10 × 10	$O(n^3)$	1,000
100 × 100	$O(n^3)$	1,000,000
1000 × 1000	$O(n^3)$	1,000,000,000

Discussion

The standard matrix multiplication algorithm has a **cubic time complexity ($O(n^3)$)**.

The number of operations increases rapidly as the matrix size grows.

It provides accurate results but becomes computationally expensive for very large matrices.

Advanced algorithms such as **Strassen's Algorithm** offer improved performance for large-scale computations.

The analysis shows that asymptotic notations are useful for evaluating and comparing algorithm efficiency.

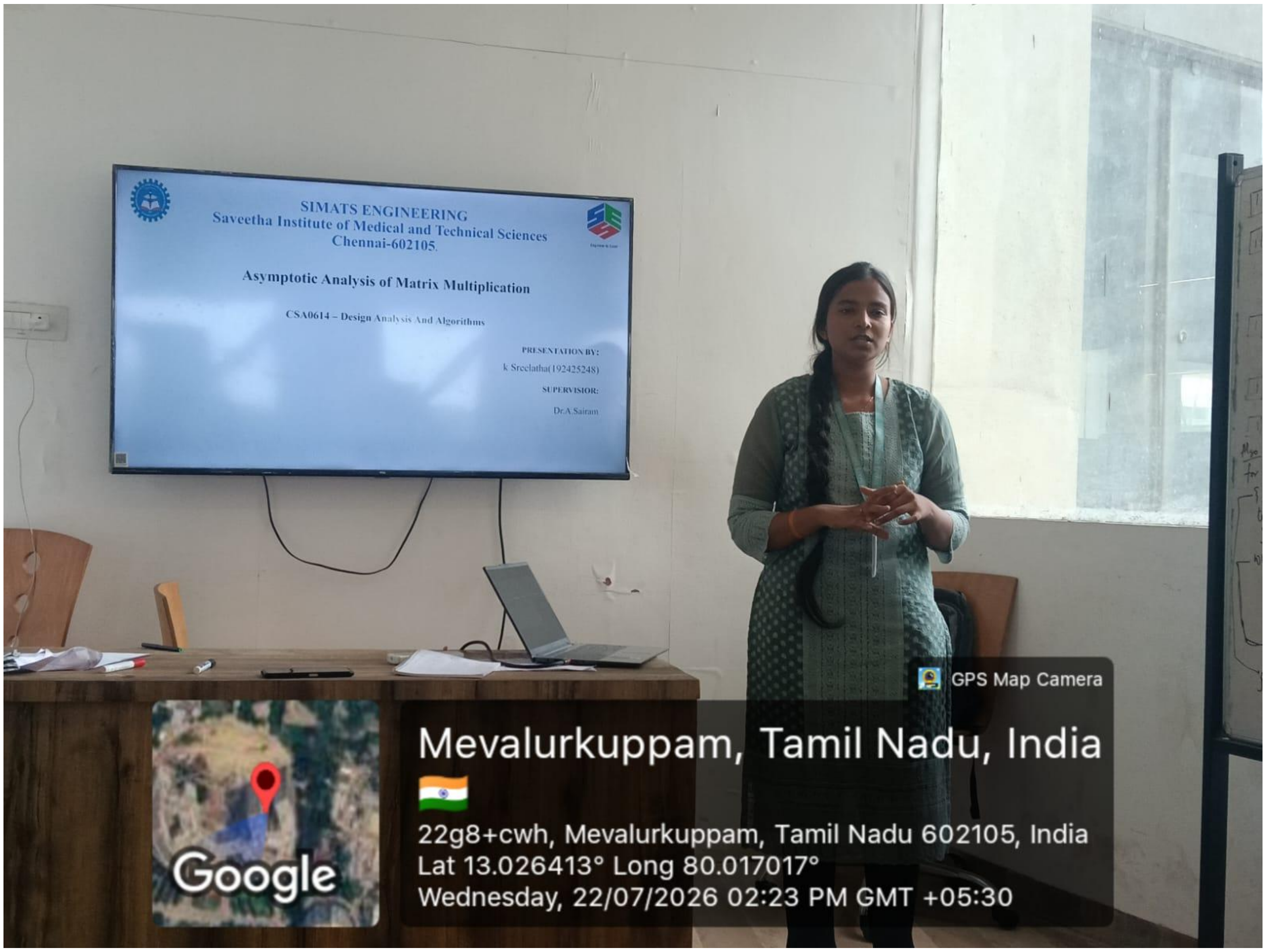
Conclusion & references

Conclusion

- Standard matrix multiplication uses **three nested loops** to compute the product of two matrices.
- The algorithm has a **time complexity of $O(n^3)$**, represented by **Big-O, Big-Theta, and Big-Omega** notations.
- As the matrix size increases, the number of operations grows **cubically**, affecting execution time.
- The standard method is simple and efficient for **small to medium-sized matrices**, while advanced algorithms are preferred for larger datasets.

References

- Thomas H. Cormen, *Introduction to Algorithms* (4th Edition).
- Ellis Horowitz, *Fundamentals of Computer Algorithms*.
- Donald E. Knuth, *The Art of Computer Programming*.
- GeeksforGeeks – Matrix Multiplication.
- NPTEL – Design and Analysis of Algorithms.


SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.
Asymptotic Analysis of Matrix Multiplication
CSA0614 – Design Analysis And Algorithms
PRESENTATION BY:
k Sreelatha(192425248)
SUPERVISIOR:
Dr.A Sairam
GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026413° Long 80.017017°
Wednesday, 22/07/2026 02:23 PM GMT +05:30
Google